

The feature interaction problem in networked multimedia services — present and future

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Feature interaction problems have been a serious issue among intelligent network services for a number of years, but, until now, the issue has not been investigated for existing and emerging networked multimedia services. This paper describes the final results of the feature interactions in a multimedia systems project which determined the level of threat that the problem posed to the networked multimedia domain. After considering the causes of the problem, some existing multimedia services are described and illustrative multimedia feature interaction scenarios provided. Issues with future networks are outlined and the paper concludes by outlining some possible approaches for handling the problem in the future.

1. Introduction

Much of the investment in telecommunications networks is now in software, with over 90% of the R&D costs in new telephone exchanges in software. Software faults have proved both costly and catastrophic — witness the recent collapse of the AT&T data network in the US. This is because a software fault, unlike a hardware fault, is replicated throughout the network. Thus software issues within telecommunications are of the utmost importance.

While telecommunications software does not have any unique characteristics to distinguish it from other domains, it does, however, have all the characteristics in the one domain — large, real-time, distributed control, distributed data, high reliability, and rapid changes. In essence, the feature interaction problem [1, 2] is caused by separate sections of the software disrupting each other, for example, call forwarding on busy interfering with the operation of call waiting. So, although it is a general problem of distributed and in particular changing software, it has been most visible within telecommunications.

The feature interactions in multimedia systems (FIMS) project has been the first to investigate the presence of the feature interaction problem in multimedia services. The 12-month project was run by the University of Strathclyde in collaboration with BT Laboratories. It was primarily conducted through paper studies and aimed to identify areas in which the problem could occur in multimedia services and ultimately provide recommendations on approaches to avoid the problem.

2. Effects of the feature interaction problem

The effect of the feature interaction problem is that features and services, which otherwise operate correctly, cannot act in harmony when placed together in a network. Indeed the service providers may not even be aware of the problem until complaints begin to filter through from their clients. By this late stage, the cost of investigating and ‘fixing’ the problem (by modifying the service) will be high in terms of labour and lost revenues from the service. Furthermore, without adopting a methodology to avoid feature interactions, it is likely that new services produced will also be subject to the same problems and maintenance costs.

Historically, the problem was resolved at the testing phase when an ‘expert’ could test the services, detect instances of the problem and manually ‘fix’ them. For example, a ‘fix’ may disallow the activation of both call forwarding and call waiting services. This happened on a per feature pair basis. However, with a rapid and continued increase in the number and complexity of services, this approach becomes unworkable. The problem is exacerbated by deregulation which will allow third party service provision and mean that no single agency will have access to or be able to test all of the services provided.

Resolving the feature interaction problem has proved to be difficult for a number of additional reasons. Firstly, the problem cannot be easily isolated to a particular set of conditions. There are examples of feature interactions

involving varying numbers of features, subscribers, components and networks. Moreover, the problem can also occur on different platforms and indeed appears to be independent of technology. All of these effects are prohibitive to rapid and cost-effective addition of networked services and features. It is essential that this problem is solved if a rapid growth of telecommunications services (and the revenues they generate) is to be realised.

3. Existing multimedia services and applications

Because multimedia applications consist of groups of services, it is possible that interactions between applications can be viewed as interactions between their component services (though the application-specific elements may also interact). It is thus important to be able to examine what services are available and identify any interactions. The multimedia services described in RACE MAGIC [3, 4] are most suitable for building applications. RACE MAGIC offered the following basic services:

- telephony — is the standard telephony service and is likely to provide most scope for interworking,
- videotelephony — is the same as telephony but also allowing exchange of video information,
- videoconference — allows real-time multi-point exchange of high quality voice and video information,
- TV distribution with selection path — allows a user to select and view a TV channel from a service provider,
- video library — provides the user with the capability to request 'movie' type information from a service provider on demand,
- telewatching — allows a user to monitor (in real time) a remote location using a remote camera,
- movie teleswitching — is a real-time multimedia three-party service which allows calling users to connect two remote items of equipment without establishing a bearer to themselves,
- connection-oriented bearer service — allows users to set up unrestricted digital information transfer over bearer connections to another party,
- video-on-demand — is a real-time multimedia retrieval service which provides a user with the facility of request programme material from a particular VoD server,
- electronic mail — provides the facility to send multimedia mail to a connectionless service which is then forwarded to an indicated addressee(s).

Nine multimedia applications were investigated in the course of the FIMS project. Although these employed a

variety of different operating platforms, terminals and networks, they could still be mapped on to the basic MAGIC services previously described. A brief description of each application follows.

- BT Ipswich Trial [5, 6] — this is a market trial, which started in the summer of 1995, of interactive television to 2500 homes in the Ipswich and Colchester area. This follows on from a technology trial in 1994. Services are ordered and controlled via a standard remote control and delivered to a normal television over copper telephone lines to most homes or fibre for the remainder.

Service elements used: video-on-demand, TV distribution with selection path, connection oriented bearer service.

- BT Inverness Experiment [7] — this trial was an experiment in teleworking in which Directory Assistance Centre-based operators worked from home. A supervisor's terminal was located in the Centre to allow the supervisor to manage and interact with the teleworkers. After 2 to 3 days training in June 1992 of the teleworkers and supervisors on the use of the teleworking system, the experiment was started and ran for one year.

Service elements used: videotelephony, electronic mail, connection-oriented bearer service.

- The Cambridge Trial [8] — this is an ATM-based trial which provides multimedia services to the home. Phase one, which started in September 1994 aimed to solve technical issues and logistics of service provision. Phase two, which started in March 1995, extended the number of users and created a 'service nursery' environment for developing and trialling new services.

Service elements used: video-on-demand, TV distribution with selection path, connection-oriented bearer service.

- ATM-based Multi-Party Conferencing System [9] — this multi-party conferencing system featured videoconferencing for up to four parties using CORBA-based middleware over ATM networks.

Service elements used: videoconference.

- Bell Communications Research Videoconferencing with Personal Presence System (PPS) [10] — the PPS is an experimental prototype for an advanced broadband teleconferencing service which allows individual user presentation control. Special multimedia bridges are used to support this service.

Service elements used: videoconference.

- Philips Telepainter [11] — Telepainter is an interactive networked multi-user game for two teams of players. The game is similar to Pictionary and allows players to communicate via audio and video connections in addition to the connections for the painting information and the control buttons.

Service elements used: videoconference.

- Telenor R&D Joint Viewing Tele-Operation Service (JVTOS) [12] — this is an application to facilitate computer-supported co-operative work in an environment with heterogeneous computers and networks sharing standard single-user applications (such as spreadsheets or desktop publishing). JVTOS allows users to share applications, intercommunicate via a picturephone and manipulate telemarkers to point to objects in shared windows.

Service elements used: videotelephony, connection-oriented bearer service.

- BT Wireplay — Wireplay provides a games network which enables users to play multiplayer games from home over the existing network using a modem. BT is currently working with game publishers to ensure that a wide range of Wireplay-compatible games is available.

Co-operating publishers include Acclaim, Activision, Electronic Arts, Interplay and Microprose. In addition to games playing, other services include registration, noticeboard, talk sessions, ladders and tournaments.

Service elements used: videotelephony.

- BT's Airline Interactive Services (ALIS) — this system allows in-flight multimedia services to be provided for passengers using the existing Skyphone or Jetphone satellite communications platform. Services include entertainment, broadcast news, shopping and hotel or car booking.

Passengers interact with the system via a specially designed screen with on-screen 'soft keys' and a handset which acts as a telephone or card reader for booking or shopping services. BT have also developed a service creation tool to facilitate service design and creation.

Service elements used: video-on-demand, telephony, TV distribution with selection path.

4. Identifying feature interaction causes in multimedia services

By examining the causes of feature interaction problems, it was hoped to be able to identify areas in networked multimedia services and applications which were susceptible to feature interaction problems. Cameron et al [1] identified three classes of problem which led to feature interaction problems in intelligent network s (IN):

- limitations on network support,
- intrinsic problems in distributed systems,
- violation of feature assumptions.

The first two classes of problem related to the actual implementation of services, covering network and distributed system issues. The third class — assumptions — represented more fundamental problems and were (in most cases) also implementation independent. They were therefore arguably the most important of the three classes and also the most difficult to detect and resolve.

This section reconsiders these causes for networked multimedia services and applications and determines if feature interactions are indeed a threat to the newer technology. Note that for investigation and analysis purposes, the MAGIC services [3, 4] were selected because they formed the 'building blocks' for all the other services.

4.1 The impact of limitations on network support

The problem of limited CPE signalling capabilities is drastically reduced by the menu-driven or graphical interfaces of most multimedia applications. However, this does not mean that when services are interworked, there will not be any problems. For example, when more than one service is simultaneously active, say video-on-demand and TV distribution with selection path, if separate 'windows' are created for both services, there is little likelihood of interactions due to limited CPE signalling capability. In this situation, there should be no confusion over which service is meant to receive a signal. However, if only one window is used or operation depends on keypresses rather than 'point and click' mouse operations, feature interaction style problems (e.g. CW+3WC) may occur.

The problem of limited functionalities for communications among network components effectively causes a race condition between features for channel bandwidth to send inter-process messages. As such, it may cause problems in multimedia systems because, like any other distributed processing system, multimedia systems will rely on inter-process communication. As Cameron et al [1] point out, these interactions bring up issues in protocol and interface design.

4.2 *The impact of problems in distributed systems*

It seemed likely that multimedia services would be affected by problems in distributed systems. Resource contentions will still be an issue with multimedia systems. Indeed, the extra bandwidth required by most multimedia services brings to the fore channel and bandwidth allocation issues which were not previously experienced in the IN world. New networks such as ATM provide bandwidth on demand, but if multimedia services based on ATM networks are interworked with services implemented on older networks (e.g. ISDN), it is possible that resource contention problems will occur. Even if dynamic bandwidth allocation is available, the user will always be limited to a certain amount of bandwidth resource which means that resource contentions may still occur.

- Personalised instantiation problems (e.g. a service instantiated with a screening list) may be an issue with multimedia systems. The problem occurred in the IN because certain sets of parameter assignments (e.g. when screened and forwarding numbers are the same) caused two otherwise independent features to interfere with each other. Such scenarios are possible in the multimedia world. For example, consider a family which subscribes to the video-on-demand service and applies a 'family lock' service on the adult videos. Services which can circumvent the 'family lock' such as multimedia-videoconference or movie tele-switching (both multi-party services) will cause interactions. Issues of billing also become a problem when multi-party services are used.
- Timing and race condition problems from the user's point-of-view should be alleviated by the improved user interface adopted by most multimedia services. The user interface should be designed in such a way that functions and controls for each service are clearly separated. For example, consider a user who has subscribed to both video-library and videotelephony services. If, while making a video-library request, the user receives a call, in order to avoid confusion as to which service is active, a window should pop up on-screen asking the user to accept or reject the call. The video-library service should therefore not be affected. If the incoming call is accepted, the video-library and videotelephony services should be presented in separate windows. A flexible operating system environment is therefore needed.
- Distributed support of features is likely to exist for multimedia systems. The major impact of this is that it makes designing feature management difficult because no single component is aware of all the service logic. 'Non-atomic operations' as outlined by Cameron et al

[1] affects service provision in IN networks. As such, it is likely that multimedia systems and services will be affected in the same manner.

4.3 *The impact of assumptions*

Because the problem of broken assumptions is, in many cases, technology independent and cannot be solved by simply adding more bandwidth resource or increasing the signalling capabilities, it is the most likely cause of feature interactions. Cameron et al [1] identified the following types of assumption — naming, data availability, administrative domain, call control, and signalling protocol. Based on this, assumptions for the MAGIC services were gleaned, and methods of breaking the assumptions were identified. The results of this analysis are shown in Table 1.

It was thus apparent from this analysis that the multimedia services and applications would be affected by feature interaction problems in much the same manner as intelligent network (IN) services. Indeed, there were even new and unique problems with limited and shared bandwidth resource. Using this information, several example scenarios were constructed to illustrate feature interaction problems in multimedia services.

5. **Feature interactions in networked multimedia**

Using the causes of feature interactions previously identified, several hypothetical scenarios were constructed to illustrate how networked multimedia services will be affected by feature interaction problems. Scenarios involving problems with 'limited network support' were not considered because it was believed that this type of problem could be eliminated by improvements to the network infrastructure and were not 'true' feature interaction problems. Note that the scenarios make use of additional non-MAGIC (telephony) services and, as such, it is assumed that it is possible to perform actions such as forwarding on video streams.

5.1 *Feature interactions due to broken assumptions*

Example 1 — Videoconference and Call Privacy (BVCS+CPS)

User A is a videoconference (BVCS) subscriber, trying to create a videoconference between users B and C. User C is a subscriber to a call privacy service (CPS), which prevents other users from obtaining the network address of the subscriber's terminal. When user A calls user C (to bring him/her into the call), user C's CPS will prevent user A's BVCS from obtaining user C's network address. Thus user C cannot be brought into the videoconference.

Table 1 Analysis of the methods of breaking assumptions.

Videoconference assumptions:	Broken by:
the controller is always in the call	features such as call-waiting switch the subscriber between calls — if the controller is also a call-waiting subscriber he/she may leave the conference to join another call
users can configure a session (through the service provider) in a flexible manner	this may not always be true — it depends on the capability of the user's terminal and the resources used by other active services
during call set-up, the following information is exchanged between users and network server: - network addresses of participating users - terminal capabilities - communications channels required - information type to be supported by the session	- this assumes a uniform addressing scheme — this may not be the case in a multi-network environment - certain features may suppress this information (e.g. call privacy, untraceable calls) - personal communication services (PCS) or mobile services may also make obtaining this information difficult
during call set-up, the server can check the following: - the other parties can accept the call - the necessary network resources are available	if the call is distributed across several switches or components, this information may be difficult to obtain
voices are summed and only the loudest speaker is heard	this suggests the use of an audio bridge — where will this bridge be situated? at the service provider? (NB RACE MAGIC describes various bridge placement configurations)
during the communication phase a user can modify a User Service Module (USM)'s quality	- this is only possible on networks which support dynamic bandwidth allocation - limited channel bandwidth prevents this - bandwidth resource contention prevents this
during the communication phase, any party can add or remove a Picture USM	- this is only possible on networks which support dynamic bandwidth or channel allocation - limited channel bandwidth prevents this - bandwidth resource contention prevents this
during the communication phase, any party can add or remove their sound USM	- this is only possible on networks which support dynamic bandwidth or channel allocation - limited channel bandwidth prevents this - bandwidth resource contention prevents this
TV distribution assumptions:	Broken by:
a TV channel consists of either: - combined and synchronised audio, video and menu data information for SDTV (standard TV) - combined and synchronised audio, video, menu data and video information for HDTV (high definition TV)	newer services may carry or manipulate a different TV channel format
TV channel information is distributed using only a distribution tree (i.e. not end-to-end user connection for each user)	services such as forwarding, distribution and conferencing services may break this assumption
Video-on-demand assumptions:	Broken by:
movie information is passed point-to-point between the user and server	services such as forwarding, distribution and conferencing services may break this assumption
Telewatching assumptions:	Broken by:
the controller allocates the first Movie USM at call set-up	resource contention problems may mean that the Movie USM cannot be set up
if the monitor and controller have different Service Module (SM) alternatives for a particular SM, the call can only be set up if a conversion is performed in the network	this assumption is broken if conversion facilities do not exist in the network
Movie teleswitching assumptions:	Broken by:
the initiator remains involved in the call until the connection should be released, then releases the call and the connection	this assumption could be broken if the initiator is interrupted by an incoming call and becomes involved in another call
Electronic mail assumptions:	Broken by:
the connectionless service is composed of one or more connection-oriented mail service (BEM-CO) actions	this assumption depends on the transport network used
the connectionless data service depends on the (connection-oriented) B-ISDN network to carry out its function. Mail is sent end-to-end, within an envelope	this assumption depends on the transport network used
the envelope may contain a package with any kind of information	this assumption depends on the transport network used
the connectionless data service will either need to use pre-assigned VPI/VCI routings or set-up and clear down connections to transmit the mail messages	this assumption depends on the transport network used

In this example, CPS breaks BVCS' assumption about the availability of data.

Example 2 — Videoconference and Call Waiting (BVCS+CW)

User A is a subscriber to videoconference (BVCS) and call-waiting (CW). (NB It is assumed that CW acts on all attempts to connect to the subscriber's terminal.) User A is the 'controller' of the conference and calls the conferees to involve them in the call (they are the 'called' users). If user A subsequently receives an incoming call from a third party, his CW service will activate. If user A then switches to the third party, the conferees in the conference call may be 'put on hold' because the CW service regards them as the called (or second) party in the call. Clearly, this would be undesirable because the conference would subsequently proceed without a controller. A solution to this problem is for the controller to transfer control of the call to one of the conferees before leaving the call.

In this example, CW breaks the assumption that the controller is always in the call and that the call ends when the controller leaves.

5.2 Feature interactions due to problems in distributed systems

Example 3 — Videoconference and Outgoing Call Screening (BVCS+OCS)

In this scenario, the interaction occurs due to distributed support of features. User A is subscribed to video-telephony, outgoing number screening (OCS), and video-conference services. The OCS service screens outgoing calls to user B. The videotelephony and OCS services (provided by company X) are provisioned on service component 1, and the videoconference service (provided by company Y) is provisioned on service component 2. It is unlikely that user A knows of this separation. This is illustrated in Fig 1.

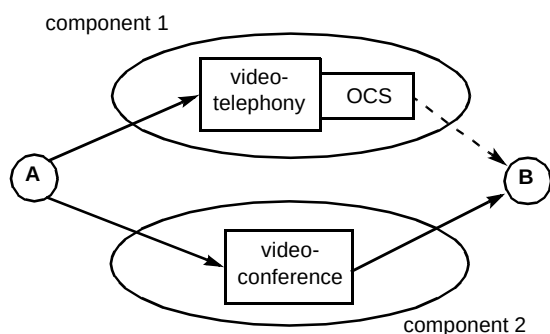


Fig 1 Videoconference + outgoing call screening.

If someone using user A's terminal attempts to make a video-telephony call to user B, they will be blocked by the screening service. However, if they use the video-conference service, they will succeed, which is clearly unintended. This interaction is caused by distributed provisioning of the services. Scenarios of this type could be a problem if user A is a parent who has screened all calls to an 'entertainment' channel (user B) and is unaware that the call screening service is inadvertently disabled when the videoconference service is used.

Example 4 — Electronic Mail + Vacation Program Service + Mailing List Service

The vacation program service (VPS) automatically replies to any received e-mails with a user-defined message. Its purpose is similar to that of an answering machine. The service is useful if the subscriber goes on holiday and wishes to inform e-mail senders of this, thus preventing his e-mail box from filling up with repeated messages from the same e-mail senders while he is away.

The mailing list service (MLS) allows users on a mailing list to mail the other users on the list by simply e-mailing the list server. If the return address for e-mails sent on the mailing list is the address of the list server (this is usual), and one of the subscribers on the mailing list enables his vacation service, the following feature interaction scenario can occur. When the vacation program receives an e-mail from the mailing list server, it will automatically send back a user-defined message to the server, thus causing the server to send it another message which the vacation program also replies to. This situation will continue until the vacation program user's e-mail box overflows or the list server is disabled.

It should be noted that restricting the list server so that it does not distribute messages to the sender of the e-mail prevents this scenario for a single vacation program subscriber, but it does not prevent the case when two or more list subscribers have activated the vacation program.

The VPS service replies 'dumbly' to any e-mails it is sent and the MLS service forwards mails 'dumbly' to all users in its mailing list. Conflicts occur when their domains of operation overlap, i.e. the VPS subscriber is also on the MLS mailing list.

Example 5 — Video-on-Demand + Video Call Forwarding (VoD+VCF)

In this scenario (see Fig 2), a billing conflict occurs through the interaction of the services. Consider a publican or bar owner who uses the video-on-demand service to provide video entertainment for clients in the bar. The owner is also a home subscriber of video-on-demand. The

charging rate of the video-on-demand service is higher for commercial users than domestic users

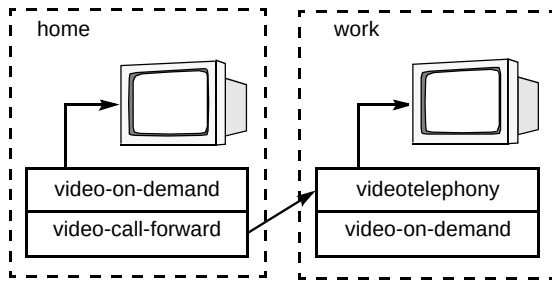


Fig 2 Video-on-demand + video-call-forward.

The owner can save money by forwarding videos from the domestic service to a terminal at work using a suitable video-forwarding service. The owner is thus charged at the (lower) domestic rate. In this scenario, the service provider loses revenue due to the feature interaction.

Example 6 — Video-on-Demand + Videoconference (VoD+BVCS)

This scenario provides an example of bandwidth resource contention. Consider a user who is subscribed to the video-on-demand and videoconference services, both of which require 2 Mbit/s to provide optimum service quality. The user's terminal is supplied with a 2 Mbit/s channel. This is illustrated in Fig 3.

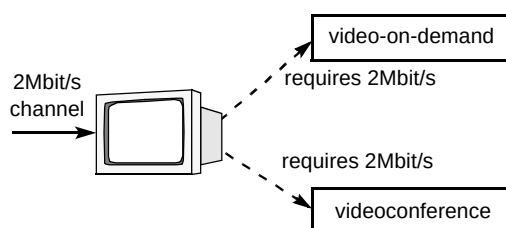


Fig 3 Video-on-demand + videoconference

If the user decides to activate one service while the other is active, either the service quality of both will deteriorate (perhaps leaving only an audio connection) or one service will not function at all. This situation is clearly unacceptable. There are many cases in which this kind of problem can occur.

5.3 Current feature interaction problems in multimedia services

The examples provided in the previous section are purely hypothetical. Currently, very little (if any) interworking exists between networked multimedia services and applications. The interworking mostly occurs between the services and the network. Therefore, most applications have to be regarded as 'stand-alone' products. The reason for this is that there does not exist a common platform from which different service providers' services may be used.

This closely parallels the personal computer world which, before the advent of windowing environments, offered little scope for users to run or interwork programs together. Now programs can share information and co-process simultaneously. That kind of flexibility (and scope for interworking) simply does not currently exist for networked multimedia services.

Even so, there exist some interaction problems between sections of multimedia applications and these are mostly detected and resolved at the specification and design stage of the application's life cycle. This *ad hoc* approach to tackling the problem will become untenable as the number and complexity of services increases.

6. The problem in future networks

This section looks to the future of networked multimedia services and examines the factors which influence the likelihood of feature interaction problems occurring. Two main factors have been identified:

- service triggering approaches,
- the concentration of intelligence.

6.1 Service triggering approaches

The degree of service triggering provided governs the degree of service interworking allowed. It is the essential factor in determining whether or not feature interactions will occur. This relationship is illustrated in Fig 4. Three reference points in the service triggering spectrum were identified and these and their effects are considered in this section.

Single-service triggering

In this extreme case, no multiple service triggering is allowed (see Fig 5). This effectively means that each subscriber can only have one service active at any one time. Although interactions between different subscribers' services can still occur, the cases when a single subscriber's services interact are eliminated.

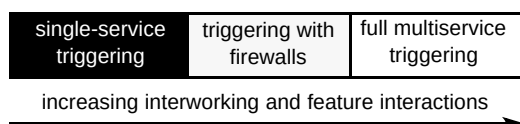


Fig 4 The service triggering spectrum.

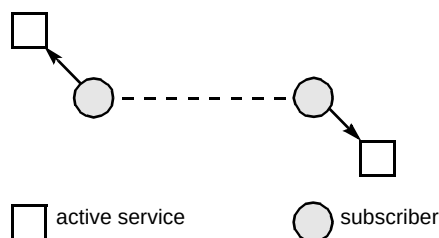


Fig 5 Single-service triggering.

Multiservice triggering — with firewalls

In this case, each subscriber can trigger multiple services but there is no possibility of feature interactions occurring due to the ‘firewalls’ built into the triggering mechanism (see Fig 6). The simplest firewall is to disallow multiple service triggering on single events. For example, if ‘call-waiting’ and ‘call-forwarding’ are active, a ‘termination_attempt’ trigger will only be presented to one of the services. Thus, services can run concurrently — for example, videotelephony service and digital clock/alarm service — while ensuring that no conflicts of intention can occur.

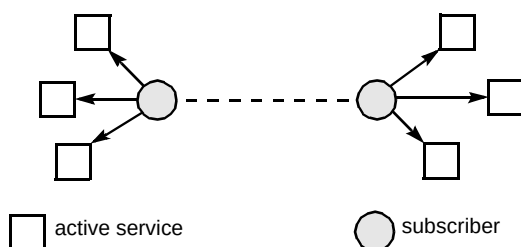


Fig 6 Multiservice triggering (with firewalls).

Although multiple services can be triggered, there is a single point of control governing service triggering (such as a basic call model in the case of telephony services). This case is analogous to the current situation with networked multimedia services and IN services. Feature interactions between the services of different subscribers can still occur in this case.

Multiservice triggering — full interworking

This case is similar to the previous except that no firewalls are used (see Fig 7). It is thus the responsibility of the feature interaction management to decide which services to trigger. This approach is useful when services share the

same trigger points but do not affect each other's operation, for example, ‘call-waiting’ and ‘incoming call screening’ services do not interact if ‘call-waiting’ has the lower priority.

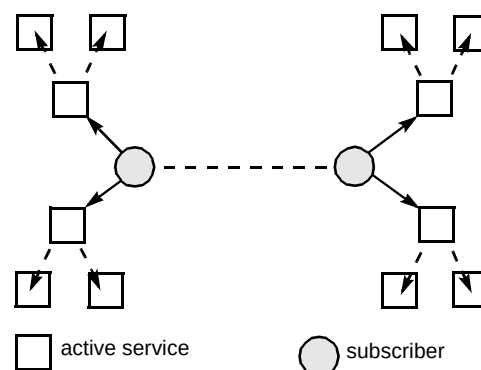


Fig 7 Full multiservice triggering.

Furthermore, services may be activated or triggered by ‘other services’. This facility is useful for services such as ‘home shopping’ in which a user may wish to automatically trigger the banking service on selecting an item to purchase. This opens up a myriad of possibilities for service interworking and for service interactions to occur.

Because there is no longer a single point from which services are activated, it may be difficult to keep track of which services are active and, consequently, feature interaction management becomes increasingly difficult. Furthermore, this level of interworking requires an operating platform which supports fully independent concurrent processes. ‘Locks’ must be placed on shared resources to avoid conflicts and increased inter-process signalling is needed to enable services to ‘negotiate’ over conflicting actions and shared resources. Managing the services becomes even more difficult if the services are implemented on a multi-component platform. Therefore, a service management approach akin to that used in distributed computer operating systems is recommended. Although potentially difficult to manage, the benefits of offering full multiservice triggering make it an attractive proposition for the future.

6.2 The concentration of intelligence

The issue of where the intelligence is to be concentrated will also have a strong effect on the potential for the feature interaction problem and the approaches required to resolve it. This can be represented by two extremes:

- intelligence concentrated in the network,
- intelligence concentrated on the periphery of the network.

Intelligence concentrated in the network

In the current public telecommunications network, the intelligence exists in the network. The user terminals and peripherals attached to the network have limited capabilities and the call-processing, service-control and service-logic execution is carried out within the network.

As witnessed in existing telecommunications networks, it is likely that feature interaction problems will occur for this case. Management of the problem must also be carried out within the network and it is likely to be centralised (to varying degrees). Centralised management is both difficult and costly. If services are provided by multiple providers, either generic management techniques will be used by the network provider, or service providers will have to supply their own service management algorithms. It is also unlikely that the user will have a significant degree of control over how interactions are resolved.

Intelligence concentrated at the periphery of the network

The other extreme of the spectrum is when the user terminals and peripherals have powerful processing capabilities and the network is used only to transport data between the terminals. Service logic and control is carried out within the terminals. The Internet is a good example of this.

In this case, the interactions will be pushed out of the network and into the periphery where they are isolated from the rest of the network. Indeed, there have been no documented examples of feature interaction problems occurring in the Internet or computer domains. The user's terminal is likely to be equipped with an operating system and user-interface environment. This allows powerful local control of the services running and may even allow the user to specify how interactions are to be resolved. In order to manage interactions which occur between different terminals or network components, advanced signalling will be required to co-ordinate the components involved in the interaction.

Although there have been no documented examples of feature interaction problems in the Internet or computer domains, this may be due to 'firewalls' which exist in these systems rather than any inherent property of concentrating intelligence in the terminals.

7. Future solutions

This section outlines some approaches to resolving feature interactions. Section 7.1 revisits the existing approaches for handling the problem and considers which ones are suitable for use in the short, medium and long term futures. Section 7.2 describes some pragmatic short-term approaches proposed by the authors which may be applied to tackle the problem in the immediate future.

7.1 Applying the existing approaches to the future

Since networked multimedia services are not radically different from their IN counterparts — control is more complex and additional media types are carried — many of the existing approaches to resolving the feature interaction problem are also applicable. This section revisits the existing feature interaction management approaches and considers which ones are applicable in the short, medium and long term futures. These approaches can be divided into several classes as illustrated in Fig 8.

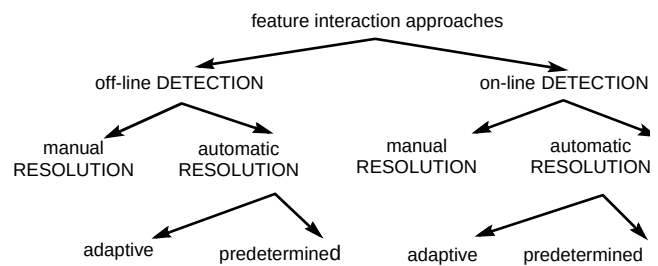


Fig 8 Classes of feature interaction approaches.

A summary of the available feature interaction management approaches and their applicability in the short, medium and long term futures is provided in Table 2.

The diverse range of specification representations poses a problem if a unified approach is to be achieved. Therefore, either a standard representation should be used for all specifications, or facilities are required to translate between different representations easily and accurately. With the long-term approaches, the feature interaction management is integral to the network and occurs on a per-interaction basis. Service providers are thus able to design, implement and provision services without worrying about the problem, leading to shorter design and testing costs.

Table 2 Feature interaction management approaches and their applicability.

Approach	Detection	Resolution	Applicability
table/rule based [13—17]	online-predetermined	automatic-predetermined	short term
behaviour analysis [18]	online-adaptive	N/A	short term
specification analysis [19—26]	offline-adaptive	manual	medium term
transactions based [27]	online-predetermined	automatic-adaptive	medium term
new architectures [28, 29]	online-adaptive	automatic-adaptive	long term
artificial intelligence [30—32]	online-adaptive	automatic-adaptive	long term

7.2 Proposed short-term resolutions

Once a feature interaction has occurred and been detected, it is then necessary to resolve it cleanly, without unnecessary disruption to the system. Multimedia applications offer improved user interfaces and can be exploited for this task. This section describes some resolution approaches devised by the authors based on this approach. The proposed resolution approaches assume the existence of a feature interaction manager (FIM) in the network.

Manual resolution

In this case, the feature interaction management prompts the user to decide the outcome of feature interaction resolutions as and when problems occur. The user will be presented with a list of possible solutions, a list dynamically generated by the feature interaction management. For example when an incoming call is detected and the following features are active — ‘call-waiting’, ‘call-forwarding’, and ‘incoming call screening’ — the user may be presented with the query shown in Fig 9.

service management message:		
!! A service conflict has occurred!!		
Please select which services to activate by checking the boxes below and then clicking on OK; click HELP for further information		
☒	video-telephony	
☐	call-forwarding	
☐	incoming call screening	
HELP	OK	call our free helpline
☒	use the same selection again next time	

Fig 9 Example status message to the user.

If the user requires further information on what the services are or would like to know the consequences of their selection, they could click on the ‘help’ button. Alternatively, the user may wish to videotelephone the service provider’s 24-hour helpline (free call) for assistance by clicking the ‘contact support service’ button. Once a selection is made, it may be stored so that when the same situation occurs again, the same action is repeated and there is no need to query the user again.

Automatic predetermined resolution

In this case, the user controls the outcome of feature interactions by presetting service priorities. Thus when service conflicts occur, the interaction is resolved by applying the predetermined service priorities. The priority list may be presented to users in the form of an on-screen table which they configure to their own requirements. Once this configuration is complete, the feature interaction management system is fully automated and requires no further user involvement.

8. Conclusions

This project has been the first to look at the feature interaction problem for multimedia services. It was clear that the problem is not a threat to current services due to the limited interworking capabilities. However, it is predicted that this will change in the future and this paper highlights issues of service triggering and concentration of network intelligence which will greatly affect the possibility of feature interaction problems. In most cases, feature interaction problems will not be caused solely by resource contentions and simply increasing the bandwidth available to services will not lessen the problem, but introduce new possibilities for ever more complex services and increased service provision.

Finally, the paper has outlined some approaches for resolving the problem once it is detected. A practical approach is proposed by the authors in addition to a review of the existing approaches and their applicability to short, medium and long term futures.

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